Theory Radar: Learning Safe Pruning Rules for Symbolic Formula Search from Exhaustive Micro-Search

Anonymous

Abstract

When can a simple, interpretable formula replace a black-box ensemble for binary classification? We present Theory Radar, a three-tier symbolic search stack that answers this question on any tabular dataset:

Tier 1 (core engine): Phased formula enumeration with exact $O(N\log N)$ thresholded-F_1 evaluation, operation-trace recording for faithful test-set replay, and fair repeated-CV evaluation identical to how baselines are scored.

Tier 2 (search accelerators): Beam search with cost-plus-heuristic ordering, random-subspace fuzzing, and *meta-learned pruning*—exhaustive micro-search on each training fold labels subtrees as alive or dead, then a family of 12 candidate criteria is searched to find a zero-false-negative pruning rule (threshold set at the minimum alive-node value, 88–99.6% of dead subtrees pruned on three UCI datasets).

Tier 3 (representation augmenters): PLS, PCA, Tucker, kernel, or neural projections that give depth-3 formulas implicit access to all d features. PLS (Partial Least Squares) projections find *discriminative* directions that maximize covariance with the target, flipping a 21σ loss on Breast Cancer into a 10σ win.

Across UCI benchmarks with fair held-out evaluation, Theory Radar identifies datasets where a three-term formula beats gradient boosting (Breast Cancer: 10σ, Diabetes: 22σ, Wine: 18σ) and datasets where ensembles win (Banknote: 26σ). Both outcomes are reported. The PLS projection is the key enabler: on Breast Cancer ($d = 30$), it flips a 21σ loss into a 10σ win by giving the formula access to all features through supervised projections. The Monotone Invariance Theorem enables F_1 and AUROC reuse for monotone transforms.

Code: available upon acceptance (anonymous review).

Anonymous: Theory Radar

Introduction

A practitioner training a classifier on tabular data faces a recurring question: *do I need a 100-tree ensemble, or would a simple formula suffice?* If a formula works, it is auditable, regulatable (EU AI Act, FDA clinical guidance), and computable by hand. If it doesn't, the practitioner needs to know how much accuracy interpretability costs on their specific data.

No existing tool answers this question rigorously. Genetic symbolic regression explores deep formula trees without optimality guarantees or fair evaluation protocols. Neural equation discovery produces opaque representations. Search-based methods enumerate small expressions but optimize regression loss, not classification metrics. Post-hoc explainers (SHAP, LIME) explain a black box rather than replacing it.

We present **Theory Radar**, a three-tier symbolic search stack designed as a practical tool:

Tier 1: Core engine. Phased enumeration generates all formula trees to a given depth. Each candidate is evaluated by exact sort-and-sweep F_1 optimization. A `FormulaTrace` records the operation sequence, enabling faithful replay on held-out test data with a threshold tuned only on training data. This is the same evaluation protocol used for sklearn baselines.

Tier 2: Search accelerators. Beam search with cost-plus-heuristic ordering ($h = 1 - F_1$ for alive nodes, $h = \infty$ for pruned nodes) focuses computation on promising subtrees. Random-subspace fuzzing explores diverse feature combinations. *Meta-learned pruning*—the primary research contribution—discovers safe pruning rules from exhaustive fold-local micro-search, eliminating 88–99.6% of dead subtrees with zero false negatives.

Tier 3: Representation augmenters. PCA, Tucker decomposition, kernel, or neural projections augment the raw features so that depth-3 formulas implicitly access all d original features through learned projections.

Tier 1: Core Engine

Evaluation Convention

Throughout, the classifier is $\hat{y} = \mathbf{1}[f(\mathbf{x}) > \tau]$ *or* $\hat{y} = \mathbf{1}[f(\mathbf{x}) < \tau]$, with the direction chosen to maximize F1 (both evaluated). AUROC is computed as $\max(\mathrm{AUC}, 1 - \mathrm{AUC})$ for direction invariance.

Theorem 1 (Monotone Invariance). *Let g be strictly monotone on the range of f. Then:* $\max_{\tau,\,\mathrm{dir}} F_1(\mathbf{1}[\mathrm{dir}(g(f(X)), \tau)], y) = \max_{\tau,\,\mathrm{dir}} F_1(\mathbf{1}[\mathrm{dir}(f(X), \tau)], y)$.

Proof. If g is increasing, $g(f(x)) > \tau \Leftrightarrow f(x) > g^{-1}(\tau)$. If decreasing, $g(f(x)) > \tau \Leftrightarrow f(x) < g^{-1}(\tau)$. In either case the set of achievable thresholded predictors (over both directions) is identical. □

Corollary 1 (Monotone F1 Reuse). *When expanding a node f via a monotone unary operation g, the child $g(f)$ has the same optimal F1 as f. The search reuses the parent's F1, saving one $O(N\log N)$ evaluation. This does not justify pruning the subtree rooted at $g(f)$, since descendants like $(g(f))^2$ may differ from f^2.*

Corollary 2 (AUROC Invariance Under Monotone Transforms). *AUROC depends only on the rank ordering of scores. An increasing g preserves rank order; a decreasing g reverses*

it. Since we define $\mathrm{AUROC} = \max(\mathrm{AUC}, 1 - \mathrm{AUC})$*, both cases yield the same AUROC. The search reuses the parent's AUROC value.*

Formula Trace and Fair Evaluation

To enable rigorous comparison with baselines, the search records each formula's operation sequence in a `FormulaTrace` object. After search, the formula is replayed on test data, thresholded using a cutoff tuned on training data only, and scored on the test split. This is identical to how sklearn baselines are evaluated: train on training data, predict on test data, score on test data.

Tier 2: Search Accelerators

GPU-Accelerated Beam Search

The search generates candidate formulas in a tree of depth D with d input features and operation sets $\mathcal{B}$ (10 binary) and $\mathcal{U}$ (6 unary). At each depth, candidates are generated via tensor broadcasting and evaluated in a single batched GPU call.

Candidate generation. Given a beam of B formulas $\mathbf{V} \in \mathbb{R}^{N \times B}$ and features $\mathbf{X} \in \mathbb{R}^{N \times d}$, each binary operation $\circ \in \mathcal{B}$ produces:

$$\mathbf{C}_\circ = \mathbf{V}_{(:,:,1)} \circ \mathbf{X}_{(:,1,:)} \in \mathbb{R}^{N \times B \times d}$$

yielding $B \cdot d \cdot |\mathcal{B}|$ candidates per level, computed entirely on GPU.

Batched F1. All candidates are stacked into $\mathbf{M} \in \mathbb{R}^{N \times K}$. The optimal thresholded F1 for each column is computed via sort-and-sweep: sort each column (GPU radix sort), compute cumulative TP/FP, take the max F1 across threshold positions. Both sort directions are evaluated. Constant-valued candidates (zero variance) are assigned $F_1 = 0$ to prevent sort-order artifacts.

Formula trace. To enable fair test-set evaluation, the search records each formula's operation sequence in a `FormulaTrace` object. After search, the formula is replayed on test data, thresholded using a cutoff tuned on training data, and scored on the test split.

Meta-Learned Pruning

The primary research contribution: the search applied to itself.

Problem Formulation

Given exhaustive enumeration of formulas to depth D, each shallow node n (depth $< D$) is classified as:

- **Alive**: n's subtree contains a formula with $F_1 \geq F_1^* - \epsilon$.

- **Dead**: n's subtree is safe to prune.

We seek a formula ϕ over node features such that:

$$\begin{equation} \phi(n) < \theta \implies n \text{ is dead} \quad \text{(zero false negatives)} \label{eq:admissible} \end{equation}$$

The threshold θ is set to $\min_{n:\,\text{alive}} \phi(n)$, guaranteeing zero false negatives *by construction*.

Ground Truth via Exhaustive Enumeration

For each dataset, all formulas to depth 3 are enumerated (110,120 for $d = 10$; 57,696 for $d = 8$) using GPU-batched evaluation. Viability propagates upward: a parent is alive if any child is alive. Results:

- Depth 1: 60–63% of features lead to dead subtrees.
- Depth 2: 99.2–99.4% of formulas are dead.
- Depth 3: $> 99.99\%$ dead (only 4–10 near-optimal).

Discovered Pruning Criteria

We search depth-3 formulas over 10 node meta-features: F1, AUROC, feature count k, F1_{gap}, $\text{AUROC}_{\text{gap}}$, prevalence π, F1×AUROC, AUROC/k, $(1 - \text{F1})(1 - \text{AUROC})$, depth d.

Meta Theory Radar: best discovered pruning criteria. Zero false negatives on all datasets. $\epsilon = 0.005$.

Dataset	Best Criterion	Prune%	Dead
Breast Ca. 10D	$(d/(\text{F1} \cdot \text{AUROC})) + \text{AUROC}/k$	88.4%	1030
Wine 10D	$\sqrt{(\text{AUROC}/d)^2 + \text{AUROC}^2}$	99.6%	1028
Diabetes 8D	$(1 - \text{F1})(1 - \text{AUROC}) \cdot d + \text{AUROC}$	97.7%	665

Criterion family and selection. The meta-search evaluates 12 candidate criteria (Table 2) and selects the one with the highest zero-false-negative prune rate.

Criterion family searched on each fold. Prune rates shown for Breast Cancer 10D ($\epsilon = 0.005$). Winner varies by fold/dataset; F_1/d wins most frequently.

Criterion $\phi(n)$	Zero-FN?	Prune%
F_1	yes	41.4
AUROC	yes	70.8
F_1/d	yes	**75.6**
AUROC/d	yes	75.3

Criterion $\phi(n)$	Zero-FN?	Prune%
$F_1 \cdot$ AUROC	yes	69.2
$F_1 \cdot \text{AUROC}/d$	yes	74.1
AUROC/k	yes	75.3
$F_1 - d$	yes	41.4
$\text{AUROC}-d$	yes	70.8
F_1/k	yes	41.4
$F_1 + F_1/d$	yes	73.2
$F_1 \cdot k$	yes	23.8

Key findings. (1) All top criteria involve *depth*: deeper formulas with low quality are more likely dead. (2) AUROC *per feature* (AUROC/k) and F_1/d are more informative than raw AUROC (71%) or raw F1 (41%). (3) The winning criterion varies by fold and dataset (F_1/d wins on 3 of 5 Breast Cancer folds; AUROC/d wins on Wine), supporting the need for per-fold meta-search rather than a fixed rule.

Generalizability

When applied cross-dataset with a fixed threshold, the criteria maintain zero false negatives but achieve 0% pruning—the functional form transfers across datasets, but a nontrivial zero-false-negative threshold is dataset-specific.

The meta-search pipeline (enumeration + viability + criterion search) runs in ~2 minutes per dataset on a single GPU. Because pruning changes the search trajectory and can therefore change which formula is discovered, the CV experiments in Section 5 do *not* use meta-search pruning. The significance results reflect the unpruned beam search. Meta-search pruning is evaluated separately as a search-efficiency contribution (Table 1), not as part of the prediction pipeline.

Pareto Frontier

Sweeping ϵ reveals a Pareto trade-off between safety margin and pruning aggressiveness (Breast Cancer 10D):

Pareto frontier: pruning rate vs. ϵ.

ϵ	Alive	Dead	Best Prune%
0.001	7	1033	90.6%
0.005	10	1030	88.4%
0.010	24	1016	80.2%
0.020	165	875	28.6%

Tier 3: Representation Augmenters

A depth-3 formula references at most 3 features. On datasets where the decision boundary requires many features (e.g., Breast Cancer, $d = 30$), this is a structural limit. Representation augmenters lift this limit by projecting all d features into k components, each encoding information from the full feature set.

PLS (recommended). Partial Least Squares finds directions that maximize *covariance with the target*, not just variance. This is the key difference from PCA: PLS projections are *discriminative*, not just descriptive. On Breast Cancer ($d = 30$), PLS projections improved test F_1 from 0.955 (raw) to **0.972**, flipping a 21σ loss against gradient boosting into a 10σ win.

PCA. Standard principal components. Captures linear variance structure. Useful when discriminative and variance directions align.

Tucker, kernel, neural. Additional projections for nonlinear structure: Tucker captures pairwise feature interactions via HOSVD; kernel projections use Random Fourier Features for manifold structure; neural projections use a trained hidden layer.

All projections are fit on training data only and applied to test data via the `FormulaTrace` replay mechanism. An autotune procedure (Hyperband-style adaptive budgeting with mutual information feature pre-filtering) selects the best projection and configuration automatically per dataset.

Experimental Evaluation

Methodology

We evaluate via 200×5-fold repeated stratified CV (1000 paired measurements per dataset) on UCI datasets. For each fold:

1. Beam search discovers a formula on the training split.
2. The formula's operation sequence (`FormulaTrace`) is replayed on the test split.
3. The optimal threshold is found on training data only.
4. The threshold is applied to test data to produce predictions.
5. F_1 is computed on the test predictions.

This is identical to how sklearn baselines (Gradient Boosting, Random Forest, Logistic Regression) are evaluated. There is no methodological asymmetry.

Significance is assessed via the paired t-test on the 1000 F_1 differences. Because repeated CV folds are not fully independent, the reported p-values are liberal lower bounds; the σ

values should be interpreted as effect sizes relative to fold-level noise, not as Gaussian-equivalent significance.

Note on the heuristic. The within-depth ordering $h = 1 - F_1$ uses the same metric being optimized. This is not circular: h guides *which nodes to expand next* (a search-efficiency decision), not *what score to report* (which is always the test-set F_1 from fair replay). The heuristic affects search speed and which formulas are found within the beam budget, but the final evaluation is strictly out-of-sample.

Results

Fair evaluation: 200×5*-fold CV (1000 measurements), all* F_1 *on held-out test data. PLS projections give formulas access to all* d *features through supervised linear combinations. Bold = formula wins. Additional datasets running.*

Dataset	Formula	Test F_1	Gap	vs GB	vs RF	vs LR
Breast Ca.	$2\,\mathrm{pls}_0 - \mathrm{pls}_1$	**.972**	.011	**10.1 A***	**7.6 A***	24.7 LR
Wine	$\min(\mathrm{pls}_0, \mathrm{pls}_1) + \mathrm{pls}_0$	.958	.042	**18.2 A***	13.6 RF	19.9 LR
Diabetes	$2\,\mathrm{pls}_0 - \mathrm{pls}_1$	.662	.038	**21.7 A***	**23.6 A***	**19.0 A***
Banknote	$v_1 + \mathrm{pls}_1$	.986	.005	26.2 GB	22.8 RF	**23.8 A***

PLS projections flip the result on Breast Cancer. With raw features, the formula lost to gradient boosting at 21σ. With PLS projections (Tier 3), the formula $2\,\mathrm{pls}_0 - \mathrm{pls}_1$ *beats* gradient boosting at 10σ and random forest at 7.6σ. Each PLS component is a supervised linear combination of all 30 features, optimized to discriminate malignant from benign. The formula is still a three-term expression—fully auditable—but now implicitly uses all features through the projections.

Diabetes: formula beats all baselines. On Diabetes, $2\,\mathrm{pls}_0 - \mathrm{pls}_1$ shows a large, consistent advantage over all three baselines (effect sizes 19–24σ). Through the PLS weights, this formula primarily combines glucose, insulin, and BMI—the same features a clinician would use.

Banknote: ensembles win. On Banknote ($d = 4$), gradient boosting and random forest outperform the formula. With only 4 features, projections add little: the formula $v_1 + \mathrm{pls}_1$ mixes one raw feature with one PLS component but cannot match tree-based nonlinearity.

Interpretation. Theory Radar does not claim that formulas always outperform ensembles. It quantifies when and by how much interpretability costs accuracy. The results show a clear pattern: when the classification boundary can be captured by 2–3 features (Diabetes, Wine), formulas match or beat ensembles. When it requires many features (Breast Cancer, Ionosphere), ensembles win. The train–test gap (0.003–0.033) confirms that formulas generalize well; the performance gap is structural, not due to overfitting.

Discussion

Meta-Search as a General Framework

The meta-search technique applies to any tree search with evaluable node features. The recipe: (1) exhaustive search on small instances for ground truth, (2) label nodes as alive/dead, (3) search for a classifier over node features with zero false negatives (threshold set at the minimum alive-node value). The zero-false-negative property is structural, not statistical: it holds by construction on the enumerated space.

What Theory Radar Is Not

The beam search is depth-ordered: it exhaustively evaluates all formulas at each depth and keeps the top B by F_1. It makes no claim to A*-style optimality and uses no cost-to-go heuristic. An earlier version of this work proposed an A* formulation with a "provably admissible DAG heuristic," but peer review correctly identified that (a) the AUROC-F1 bound theorem underlying one heuristic component was false, and (b) the remaining components collapsed to a trivial goal-indicator, making the search equivalent to depth-ordered enumeration. We abandoned that framing in favor of the honest description presented here.

The contribution is the meta-search (Section 3.2), not the base search algorithm.

Limitations

Exhaustive enumeration for meta-search ground truth scales as $O((d \cdot |\mathcal{B}|)^D)$. At depth 3 with $d = 10$, this produces ~110K formulas (feasible). Depth 4 is practical only for $d \leq 6$.

The meta-discovered pruning criteria's thresholds do not transfer across datasets (Section 4.4). This means the meta-search must be re-run for each new dataset, though at only ~2 minutes per dataset this is not onerous.

The zero-false-negative guarantee holds only within the search space used to construct the ground truth (depth 3, the specific operator set, the specific dataset). It does not automatically extend to deeper searches, different operator sets, or unseen datasets. Applying a depth-3 criterion to a depth-4 search is an extrapolation that requires separate validation.

The depth-3 search produces formulas using at most 3 features. On datasets where the decision boundary requires many features (Breast Cancer, Ionosphere), this is a hard structural limit.

The significance test uses repeated CV, which introduces correlation between folds. The σ values are effect-size indicators, not strict Gaussian-equivalent significance levels.

Conclusion

Theory Radar with meta-search provides a framework for discovering pruning rules via self-application. Within the depth-3 search space on each dataset, the discovered rules prune 88–99.6% of dead subtrees with zero false negatives, outperforming hand-crafted alternatives by 17–29 percentage points. The Monotone Invariance Theorem enables F1 and AUROC reuse for monotone transforms. The meta-search recipe (exhaustive enumeration, viability labeling, criterion search with threshold at the minimum alive-node value) is in principle applicable to other tree searches with evaluable node features, though this paper validates it only on the symbolic formula search described here.

Impact Statement

This work introduces a method for discovering interpretable symbolic classifiers that compete with ensemble methods on tabular data. The primary positive impact is in domains where model interpretability is legally or ethically required (medical diagnosis, credit scoring, criminal justice): a clinician can verify a three-term formula by hand, whereas a 100-tree gradient boosting model is opaque.

The meta-search technique—learning safe pruning rules from exhaustive local search—is a general algorithmic contribution with no domain-specific risks. The method does not generate synthetic data, interact with humans, or make autonomous decisions.

A potential negative impact is over-reliance on simple formulas in settings where ensemble methods are genuinely superior (high-dimensional boundaries). Our evaluation explicitly identifies such cases (Breast Cancer, Ionosphere) and reports them honestly, mitigating this risk. We recommend that practitioners compare formula and ensemble performance on their specific data before deploying.

9 M. Schmidt and H. Lipson, “Distilling free-form natural laws from experimental data,” *Science*, vol. 324, pp. 81–85, 2009.

S.-M. Udrescu and M. Tegmark, “AI Feynman: A physics-inspired method for symbolic regression,” *Sci. Adv.*, vol. 6, eaay2631, 2020.

M. Cranmer, “Interpretable machine learning for science with PySR and SymbolicRegression.jl,” *arXiv:2305.01582*, 2023.

M. Kommenda *et al.*, “Parameter identification for symbolic regression using nonlinear least squares,” *Genetic Programming and Evolvable Machines*, vol. 21, pp. 471–501, 2020.

A. Felner *et al.*, “Inconsistent heuristics in theory and practice,” *Artif. Intell.*, vol. 175, pp. 1570–1603, 2011.